

Abdelrhman Elsaid

Software Engineer | .NET Backend Developer

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Summary

.NET Backend Developer with hands-on experience building RESTful APIs using C#, ASP.NET Core, Entity Framework Core, and SQL Server. Experienced in designing layered backend systems using **Onion Architecture**, **N-Tier Architecture**, CQRS with MediatR, Repository Pattern, JWT authentication, role-based authorization, and unit testing. Built practical Web API projects including a Hotel Reservation System and an Examination System with authentication, business workflows, data validation, automated grading, and 100+ passing unit tests.

Education

Bachelor of Engineering in Electronics and Communications Engineering Mansoura University Egypt 2020 – 2025

Activities and Societies:

- **Multimedia Director | Luminous MU** Jan 2023 – Jul 2024 [Sessions and Workshops](#)
Recruited and interviewed over **30 new members**, restructuring the media and marketing team to improve operational efficiency.
 - **Sub-team Leader | IEEE Mansoura Student Branch** Jan 2022 – Dec 2023 [Certificate](#)
Managed the end-to-end creation of visual assets for 2 major events as the design sub-team lead, ensuring timely delivery and brand consistency.
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Technical Skills

Languages: C#, C++, SQL

Backend Development: ASP.NET Core Web API, ASP.NET MVC, RESTful APIs, LINQ, ADO.NET

Databases & ORM: SQL Server, Entity Framework Core, Code-First Migrations

Architecture & Patterns: OOP, SOLID Principles, Design Patterns, Onion Architecture, 3-Tier Architecture, Repository Pattern, CQRS, MediatR, Fluentvalidation

Authentication & Security: JWT Authentication, Refresh Tokens, Role-Based Authorization, Network Security Fundamentals

Testing & Logging: Unit Testing, NUnit, Moq, FluentAssertions, serilog, seq

Tools: Git, GitHub, Visual Studio, Postman, Swagger, Trello

AI & Prompt Engineering: LLM API Integration, Structured Outputs, Prompt Engineering

Projects

- **Hotel Reservation System Web API** May 2026 [Project Link](#)
 - Tech Stack: Web API | ASP.NET Core, EF Core, SQL Server, CQRS, MediatR, JWT.
 - Developed a RESTful API for hotel reservation management using **Onion Architecture** and **CQRS** with MediatR.
 - Implemented **JWT** authentication, **refresh tokens**, **custom authorization filters**, and centralized error handling.
 - Added **transaction middleware** to ensure data integrity across reservation and room management workflows.
 - Used EF Core, **Repository Pattern**, and AutoMapper for clean data access and **DTO mapping**.
 - Created 82 passing **unit tests** with NUnit, Moq, and FluentAssertions covering handlers and **orchestrators**.
- **Examination System Web API** Apr 2026 [Project Link](#)
 - Tech Stack: Web API | ASP.NET Core, EF Core, SQL Server, Identity, JWT, AutoMapper, **FluentValidation**.
 - Developed a RESTful API for managing online exams, courses, questions, students, instructors, submissions, and grades.
 - Implemented **JWT authentication** and role-based authorization using ASP.NET Core **Identity** for Admin, Instructor, and Student roles.
 - Built random exam generation and automated grading, Used N-Tier Architecture, **Repository Pattern**, and AutoMapper to separate API, business logic, data access, and **DTO mapping**.
 - Created **32 passing unit tests** with NUnit, Moq, and FluentAssertions covering core business services and implemented **centralized logging** using Serilog and Seq to enhance system observability.

Certificates

Practical Prompt Engineering [Frontend Masters] Certificate	Apr 2026
Backend Development Foundations (C# & SQL) [Programming Advice] Certificate	Oct 2025
Computer Science & Problem Solving (C++) [Programming Advice] Certificate	Feb 2025
Huawei Certified ICT Associate – Datacom Certificate	Aug 2024

Languages

Arabic: Native

English: intermediate